

Toward Measuring Prompt Quality: A Preliminary Investigation on Prompt Smells

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Abstract—Large Language Models (LLMs) are increasingly embedded in software engineering workflows, yet the quality of the prompts used to guide these models remains largely unassessed. Poorly formulated prompts can lead to ambiguity, degraded output quality, or biased and unsafe model behavior. In this paper, we conduct a preliminary exploratory study aimed at establishing an initial foundation for systematic prompt-quality evaluation. First, we perform a rapid review of the literature on prompt engineering and analyze real-world prompts to derive a preliminary taxonomy of *prompt smells*—recurring patterns that signal potential quality issues. Each smell is operationalized through a quantitative metric capturing aspects such as grammatical correctness, readability, contextual adequacy, complexity–length, formality mismatch, and bias.

We then apply these metrics to an existing corpus of developer-written prompts and analyze their relationships using correlation analysis and multiple regression models. Our results show that syntactic metrics (e.g., grammar, readability) are strongly interrelated and largely explain the composite Prompt Quality Score, whereas higher-level smells such as contextual relevance, formality mismatch, and bias exhibit weak correlations and low explained variance. These findings suggest that many important prompt-quality issues operate independently of surface-level text properties and may require more semantically informed assessment techniques. Overall, this study offers an initial conceptual and empirical basis for the systematic evaluation of prompt quality, paving the way for more robust prompt engineering practices and automated quality-checking tools.

Index Terms—Prompt Quality; Prompt Smells

I. INTRODUCTION

LLMs are nowadays widely adopted to support natural language natural language and software engineering (SE) tasks, mostly without task-specific fine-tuning [1]. In this setting, *prompts*, i.e., the natural language instructions provided as a query to the model to guide it towards a precise answer [2], are the main artifact used by practitioners to control models’ responses: instead of changing model parameters, users write prompts to instruct the model on what to do, how to do it, and in which style [2]. This has led the SE community to introduce a new discipline, i.e., *prompt engineering*, that investigates specifically *how* to engineer prompts to guide LLMs in reliably providing qualitative answers. Recent surveys list many prompt engineering strategies, and show that the way a task is written in the prompt can be as important as the choice of model itself [3].

Although prompt engineering is now common, most academic work evaluates LLMs based on their task outputs, rather

than on the intrinsic **qualitative properties of the prompts as SE artifacts**. Surveys on LLM evaluation organize the literature around evaluation tasks, benchmarks, and metrics like accuracy, BLEU, ROUGE, F1, human judgments, or safety scores, and treat the prompt as part of the experimental setup rather than as an object of evaluation on its own [4], [5]. A recent review by Joshi et al. [6] reaches a similar conclusion and summarizes LLM evaluation primarily in terms of task-level and system-level metrics, while only briefly noting emerging work on prompt-level metrics. There is still no shared, multi-dimensional model of *prompt quality* that can be computed automatically and studied across datasets.

In common SE research, a related problem is addressed through the concept of *code smells* [7]. These are surface-level indicators in source code that often point to deeper design problems, such as very long methods, large classes, duplicated code, or overly long parameter lists. Over time, catalogs of code smells have been created and integrated into tools to guide refactoring and improve maintainability [8]. A code smell is not a bug; it is a warning that a piece of code might be hard to understand, modify, or test in the future. *In this paper, we position prompts as the artifact that engineers use to guide LLMs in providing reliable and high-quality responses, and investigate intrinsic qualitative aspects of such prompts.* Many low-quality prompts show obvious surface issues, for example, “*Explain this*” with no context, or “*Give me a perfect answer, you must be 100 percent correct*”, with no task description. Other prompts include unclear pronouns (e.g., “*Do it like before*”), inconsistent tone, or biased framing (e.g., “*Explain why group X is worse than group Y*”). These are not syntax errors, but they are signs that the prompt may lead to brittle, confusing, or unsafe model behavior. We refer to such recurring surface issues as ***prompt smells***.

However, in the current landscape of *quality assurance for LLMs*, there is no consolidated taxonomy of prompt smells that combines insights from the literature with observations from real prompts. Furthermore, there is a lack of operational metrics that map different dimensions of these quality issues to concrete quantitative indicators, such as grammatical quality, structural complexity, contextual adequacy, style, and bias. Finally, there is very limited empirical work on how such metrics relate to one another across realistic prompt corpora, to understand what characterizes prompt quality.

Research Gap and Objective

While prompts are commonly used as the main SE artifact to interact with LLMs, there is a lack of definitions and measurements of *prompt smells*, which are hidden concerns in prompts that may lead to worse results from the models. To address this gap, we provide an initial catalog of prompt smells and quantify these concerns using a set of metrics.

To this end, our study is organized into three main phases. First, we propose a taxonomy of prompt smells, based on a rapid review of the literature on prompt engineering and software engineering, as well as an exploratory analysis of real prompts. Second, we define and implement a set of automatic metrics that act as operational proxies for these smell categories. Third, we apply these metrics to a dataset of prompts written by software developers used in prior work on prompt patterns and code quality [9]. We examine how the metrics behave and their relationships using descriptive statistics, correlation analysis, and regression models.

Our study produced two main findings. First, by synthesizing insights from the literature and examining real-world prompts, we derived a preliminary catalog of prompt smells capturing recurrent quality issues in prompt formulation. These smells span grammatical correctness, formatting and structure, readability, contextual adequacy, complexity-length, formality mismatch, and bias-related concerns. Each smell was operationalized through a corresponding metric, enabling quantitative assessment across large prompt corpora.

Second, our empirical analysis revealed that the smells do not behave uniformly. Basic linguistic indicators (grammar, readability, formatting) are strongly interrelated and collectively explain nearly all variance in the composite Prompt Quality Score. In contrast, higher-level smells, namely context relevance, complexity-length, formality mismatch, and bias, demonstrate weak correlations with syntactic features and exhibit substantially lower explained variance in regression models. As such, these results indicate that many meaningful prompt quality issues operate independently of surface-level text properties, highlighting the need for richer semantic metrics to capture the full spectrum of prompt smells.

Contributions. Our contributions to the state of the art on *prompt quality* present in the replication package [10], and summarized in the following:

- A preliminary taxonomy of *prompt smells*, grounded in a rapid review of the literature and an analysis of real prompt examples, which systematizes common quality issues in prompt formulation.
- An operationalization of each smell through a corresponding metric, providing a first quantitative framework for characterizing prompt quality dimensions at scale.
- An empirical analysis how these metrics relate to one another in practice using correlation and regression models over a real-world prompt dataset, revealing which

dimensions co-vary and which capture orthogonal aspects of prompt quality.

Together, these contributions offer an initial conceptual and empirical foundation for more systematic, measurable, and automatable evaluation of prompt quality in software engineering and AI-assisted development settings.

II. BACKGROUND AND RELATED WORK

Our work builds on the insights from several studies in the realm of Large Language Models for software engineering.

In the context of LLM usage, a prompt is defined as a “*set of instructions or input data provided to a Large Language Model to guide its output*” [11]. Effective prompt design has been shown to directly shape the model’s generated responses. For instance, Brown et al. [12] demonstrated that *prompt engineering*, that is, the practice of crafting precise inputs to improve the capabilities of LLMs [13], may sometimes yield results comparable to or even better than model fine-tuning, which requires extensive task-specific training data to adjust the model’s pre-trained weights. Indeed, prompt engineering leverages the model’s existing knowledge, allowing for optimized performance without the need for additional training.

Sasaki et al. [14] further defined *prompt engineering patterns* as “*a systematic approach to structuring interactions, providing a versatile framework applicable across various domains*”. The significance of prompt engineering in software engineering has been pointed out by White et al. [15], who found that effective prompt usage can enhance early stages of the software development lifecycle. Similar results were found by Arvidsson and Axell [16] and Rodriguez et al. [17], who assessed the role of prompt engineering on requirements engineering and traceability recovery tasks, respectively.

The notion of prompt smells has been introduced to describe semantic and syntactic issues in prompts that reduce the desirability, explainability, or traceability of generative AI outputs, drawing a direct analogy to code smells. Ronanki et al. define prompt smells as imprecise prompt characteristics that correlate with undesirable outputs [18]. Building on the long line of work on requirements smells, Femmer has proposed a frame of smells as recurring symptoms of quality defects in natural language requirements [19], which later work has refined and operationalized for automated analysis [20]. More recently, Vogelsang et al. have studied how requirements smells affect LLM-based requirements-code traceability, explicitly treating prompts as expressions of requirements and showing that the presence of smells can measurably influence LLM performance in some tasks [21].

Compared to existing work on prompt smells and requirements smells, this article extends the state of the art in three main ways. First, it proposes a taxonomy of prompt smells that is explicitly grounded both in literature findings and in an analysis of real-world developer prompts, rather than relying solely on conceptual examples. Second, it operationalizes these smells through a set of quantitative metrics that combine static text analysis and LLM-based judgments, making prompt

quality observable and measurable. Third, it empirically studies the relationships between these metrics and the identified smell dimensions using correlation and regression analyses, providing initial evidence on which aspects of prompt quality can be captured by automatic indicators and where current metrics remain insufficient.

III. RESEARCH METHOD

In this section, we present the methodological approach adopted to construct a preliminary taxonomy of prompt smells and to empirically investigate their measurable characteristics. Our method combines (i) an exploratory phase, aimed at identifying and defining prompt-smell categories, and (ii) an analytical phase, focused on understanding how measurable indicators of these smells relate to one another in practice.

A. Research Questions

To guide our investigation, we formulated the following research questions:

RQ₁ – *What types of prompt smells can be identified?*

RQ₁ seeks to uncover and categorize recurring issues that negatively affect prompt quality. These issues may include missing context, grammatical flaws, unnecessary verbosity, stylistic inconsistencies, or biased and unsafe formulations. Understanding these categories is essential to establishing a shared vocabulary for discussing prompt defects and to supporting the development of automated quality-assessment tools. To answer this question, we conducted a rapid review of the literature on prompt engineering and software engineering quality, complemented by an inspection of real-world prompts. This combined approach allowed us to derive a working taxonomy grounded both in conceptual knowledge and in practical examples.

RQ₂ – *How do the prompt smells metrics relate to each other in practice?*

RQ₂ investigates the empirical relationships among the different dimensions of prompt quality identified in **RQ₁**. While each prompt smell represents a distinct conceptual construct, their manifestations in real prompts may not be independent. For instance, prompts that are overly long may also exhibit reduced readability; prompts with missing context might correlate with vague phrasing or structural inconsistencies. This research question aims to determine whether the extracted metrics capture complementary or overlapping aspects of prompt quality, whether certain smells tend to co-occur, and whether some dimensions may be redundant. To answer this question, we analyze the metrics through pairwise correlations and multiple regression models to reveal patterns of association and to assess their collective explanatory power.

The research method applied is summarized in Figure 1. Specifically, we performed a rapid literature review to identify prompt-design issues, validated and refined these issues

by inspecting real-world prompts, consolidated them into a catalog of prompt smells and corresponding metrics, extracted and quantified these smells over a developer-prompt dataset, and finally analyzed correlations and predictive relationships among the metrics through statistical modeling.

B. Data Collection and Analysis

To construct our taxonomy and address **RQ₁**, we carried out a *rapid literature review* [10] focusing on prompt engineering, prompt design, and software engineering quality. We opted for a rapid review rather than a full systematic review because the domain is evolving at a very fast pace, and the volume of potentially relevant studies would render a systematic protocol impractical in terms of screening effort and data extraction. Our objective at this stage is to derive an initial, structured set of prompt smell categories rather than to produce a comprehensive state of the art synthesis. Consequently, the additional time and resources required for a full systematic review would not provide proportional benefits for this exploratory investigation.

This methodological choice aligns with the early-stage nature of our work, and we explicitly acknowledge that the resulting taxonomy is preliminary and intended to be refined and extended in future research. To identify relevant literature, we searched four widely used scholarly databases: *Google Scholar*, *Scopus*, *IEEE Xplore*, and the *ACM Digital Library*.

We included studies that (i) examined how prompts are formulated or structured, or (ii) introduced criteria, patterns, or guidelines describing desirable or undesirable prompt properties, such as clarity, completeness, bias, or safety. We excluded vision papers that lacked concrete prompt examples, as well as benchmark-oriented papers that focused solely on accuracy or functional correctness without any discussion of prompt design aspects.

To complement the insights from the literature and gain a clearer understanding of how prompt quality issues manifest in practice, we additionally inspected prompts drawn from a corpus of real-world developer-written prompts retrieved from GitHub [9]. This qualitative examination allowed us to observe recurring problems such as missing task context, overly long or convoluted instructions, and biased or unsafe formulations.

We then consolidated the issues identified across both sources, i.e., the academic literature and the prompt corpus, into an initial set of prompt-smell categories. Each category captures a distinct dimension of prompt quality. This taxonomy is intentionally preliminary, serving as a structured foundation for subsequent metric definition rather than as a final or exhaustive catalog of prompt quality problems.

To address **RQ₂**, we first operationalized the prompt-smell categories identified in the literature by defining measurable indicators for each smell. Based on these definitions, we implemented a set of extraction procedures using two Python libraries: *TEXTSTAT*, to compute readability indices and lexical features, and *LANGUAGE TOOL*, to detect grammatical, stylistic, and surface-level issues. These tools allowed us to

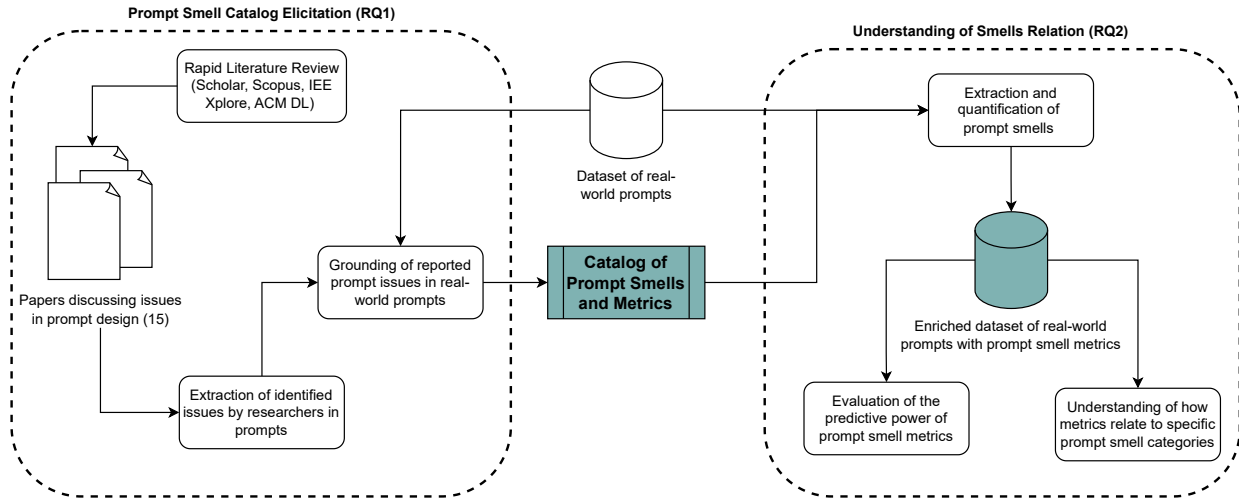


Fig. 1: Overview of the research method applied.

translate each smell dimension into a concrete metric or set of metrics that reflect the aspects identified in the literature.

We then applied these extraction procedures to all prompts contained in the dataset of real-world prompts introduced in prior work [9]. This step enabled us to quantify the presence and intensity of prompt smells in existing developer-written prompts and to obtain a structured metrics dataset for subsequent analysis.

Using this dataset, we computed a Pearson correlation matrix to explore linear relationships among all variables. Finally, we fitted a series of multiple ordinary least squares (OLS) regression models—each using one smell-related dimension as the dependent variable—to evaluate the joint predictive power of the extracted metrics and to investigate how different features collectively relate to specific prompt-smell categories.

IV. ANALYSIS OF THE RESULTS

In this section, we elaborate on the results of our preliminary study for the two **RQs**.

A. **RQ₁** — A Catalog of Prompt Smells

To address **RQ₁**, we first carried out a rapid review of the literature on prompt engineering, prompt quality, and automatic text analysis. The search was conducted on Google Scholar, chosen for its broad disciplinary coverage and inclusion of preprints and technical reports. The search produced 30 articles in total; after applying inclusion criteria (discussion of prompt formulation, quality, or evaluation, with at least some explicit criteria or metrics) and excluding purely divulgative, non-pertinent, or inaccessible works, 15 papers were retained for qualitative analysis. The screening was performed independently by two researchers, who then compared and reconciled their selections to reduce selection bias. From these 15 studies we extracted recurring notions of “good” and “bad” prompts, which we then organized into the prompt-smell categories used in our taxonomy presented in Table I.

a) Grammar and Surface-Form Smell: This smell captures linguistic defects such as misspellings, incorrect grammar, and malformed sentences, which hinder clarity and may lead to unintended model interpretations. It is operationalized through the *Grammatical Correctness* metric (G), which quantifies the density of grammar and spelling issues detected by LANGUAGE TOOL.

b) Formatting and Structural Smell: Poor formatting—including inconsistent punctuation, capitalization, or layout—can make prompts harder to parse and obscure task structure. The *Formatting Score* (F) reflects this dimension by aggregating structural indicators into a normalized quality measure.

c) Readability Smell: Prompts that are overly complex, dense, or ambiguous reduce comprehensibility and may impair the model’s instruction-following behavior. The *Readability Score* (C) captures this smell via standard readability indices measuring textual accessibility.

d) Missing or Insufficient Context Smell: A frequent issue in both literature and practice is the absence of key task information (e.g., missing objective, domain constraints, or data description). The *Relevance Context Score* (RCS) reflects this smell through an LLM-based adequacy evaluation of whether the prompt provides sufficient explicit context.

e) Over-Complexity and Excessive Length Smell: Prompts with unnecessary verbosity, compound instructions, or excessive length may confuse the model or dilute essential information. The *Complexity-Length Score* (CLS) operationalizes this smell using word count and the Gunning Fog Index to capture structural and cognitive complexity.

f) Style or Formality Mismatch Smell: Several studies highlight mismatches between the style of the prompt and the expected style of the answer (e.g., informal prompt requesting a formal output). The *Formality Mismatch Score* (FMS) measures the deviation between the prompt’s estimated formality and the intended target formality.

g) Bias and Unsafe Framing Smell: Prompts may contain biased, unsafe, or discriminatory language that can induce

TABLE I: Results of **RQ₁**: a catalog of prompt smells and metrics.

Prompt Smell	Scale	Metric Computation (Scores)
Prompt Quality (PQS)	Continuous, [0, 1]	$PQS = \frac{G + F + C}{3}$ Where G : grammatical correctness; F : formatting quality; C : readability.
Grammatical correctness (G)	Continuous, [0, 1]	$G = 1 - \frac{n_{\text{matches}}}{\max(1, n_{\text{words}})}$ Where n_{matches} : grammar / spelling issues; n_{words} : words.
Formatting (F)	Continuous, [0, 1]	Normalised combination of punctuation, capitalisation and layout indicators.
Readability (C)	Continuous, [0, 1]	Readability metrics
Relevance Context (RC)	Continuous, [0, 1]	$RCS = \sigma(s_{\text{llm}})$ $s_{\sigma(\text{llm})}$: LLM score for adequacy of explicit context.
Complexity-Length (CL)	Continuous, [0, 1]	$CLS = 1 - \min\left(1, \frac{\frac{WC}{WC_{\text{max}}} + \frac{GFI}{20}}{2}\right)$ WC : word count; WC_{max} : length threshold; GFI : Gunning Fog Index.
Formality Mismatch (FM)	Continuous, [0, 1]	$FMS = s_{\text{formal}} - t_{\text{formal}} $ s_{formal} : estimated prompt formality; t_{formal} : desired formality.
Bias Detection (BD)	Continuous, [0, 1]	$BDS = \alpha b_{\text{lex}} + (1 - \alpha) b_{\text{llm}}, \quad \alpha \in [0, 1]$ b_{lex} : lexical bias indicator; b_{llm} : LLM bias score.

TABLE II: Pearson correlation matrix among the analyzed metrics.

	PQS	G	F	C	RCS	CLS	FMS	BDS
PQS	1.000	0.793	0.545	0.677	0.068	0.260	-0.245	0.132
G	0.793	1.000	0.134	0.534	0.029	0.258	-0.178	0.097
F	0.545	0.134	1.000	-0.131	0.266	-0.323	-0.208	0.069
C	0.677	0.534	-0.131	1.000	-0.180	0.632	-0.100	0.098
RCS	0.068	0.029	0.266	-0.180	1.000	-0.299	-0.170	-0.043
CLS	0.260	0.258	-0.323	0.632	-0.299	1.000	-0.001	0.077
FMS	-0.245	-0.178	-0.208	-0.100	-0.170	-0.001	1.000	-0.089
BDS	0.132	0.097	0.069	0.098	-0.043	0.077	-0.089	1.000

harmful model behaviors. The *Bias Detection Score* (BDS) combines lexical bias indicators and LLM-based semantic bias assessments to quantify the degree of unsafe or discriminatory framing.

h) Overall Prompt Quality Smell: While not a smell category per se, several studies propose composite indicators of overall linguistic and structural quality. The *Prompt Quality Score* (PQS) synthesizes grammatical correctness, formatting quality, and readability into a single metric reflecting baseline prompt quality.

B. **RQ₂** — Relations Between Prompt Smells

To answer **RQ₂**, we first inspected the Pearson correlation matrix reported in Table II. As expected, *PQS* shows strong positive correlations with the basic syntactic metrics G and C (0.793 and 0.677, respectively) and a moderate correlation with F (0.545). This pattern reflects the fact that *PQS* is constructed from G , F , and C , so its association with them is largely built in. The three base metrics also display non-trivial

TABLE III: Coefficient of determination (R^2) for the OLS models fitted for each target metric.

Prompt Smell	R^2
Prompt Quality Score (PQS)	1.00000
Grammatical correctness (G)	0.99999
Formatting score (F)	0.99999
Readability score (C)	0.99999
Relevance Context Score (RCS)	0.14744
Complexity-Length Score (CLS)	0.47811
Formality Mismatch Score (FMS)	0.09162
Bias Detection Score (BDS)	0.03161

relationships among themselves: G and C correlate moderately (0.534). At the same time, F is only weakly correlated with G (0.134) and slightly negatively correlated with C (-0.131), suggesting that readability (F) captures a somewhat different aspect of prompt quality than grammatical correctness and formatting.

The higher-level metrics show a more articulated picture. *CLS* correlates positively with C (0.632) and, to a lesser extent, with *PQS* (0.260) and G (0.258), and negatively with F (-0.323) and *RCS* (-0.299). This indicates that prompts with more complex or penalizing complexity-length profiles tend to be less readable and less contextually grounded. *FMS* exhibits small negative correlations with most other metrics, especially *PQS* (-0.245), F (-0.208), and G (-0.178), suggesting that formality mismatch is only weakly related to the basic linguistic quality of the prompt. *BDS* is nearly uncorrelated with the other dimensions, with coefficients close to zero in every case, which already suggests that bias-related issues are largely orthogonal to the textual signals captured by the current

metrics.

To quantify the overall predictive power of the metrics, we then fitted multiple OLS regression models, each using all remaining metrics as predictors for a single target (one target at a time). The resulting coefficients of determination are reported in Table III. For *PQS* and the three base metrics *G*, *F*, and *C*, the models achieve an almost perfect fit, with R^2 values of 1 or very close to 1. This reflects the strong linear dependencies already visible in the correlation matrix and confirms that these four metrics are essentially different parametrizations of the same underlying syntactic signal. In other words, when the set of predictors includes the remaining syntactic indices, both *PQS* and *G/F/C* become almost perfectly predictable.

The situation is very different for the higher-level metrics. *CLS* attains a moderate $R^2 \approx 0.48$, indicating that approximately half of its variance is explained by linear combinations of the other metrics. Within this model, *RCS* emerges as a significant negative predictor (with a negative coefficient and a small p -value), suggesting that prompts that are more contextually relevant tend to exhibit less penalizing complexity–length profiles. For *RCS* itself, the model achieves only $R^2 \approx 0.15$, with most coefficients not statistically significant, indicating that contextual relevance is driven by semantic and pragmatic information that cannot be reconstructed from the current syntactic indicators.

The explanatory power of the models is even lower for *FMS* and *BDS*. For *FMS*, the R^2 is approximately 0.09, with *RCS* again serving as a significant negative predictor: prompts with higher contextual relevance tend to exhibit a smaller mismatch between the expected and actual levels of formality. However, the overall variance explained remains limited. For *BDS*, the R^2 drops to approximately 0.03, and no predictors exhibit meaningful effects, implying that bias-related issues are nearly completely independent of the linguistic and structural metrics considered here.

V. DISCUSSION

Our results highlight two complementary dimensions of prompt quality: (i) a set of *syntactic smells*, which capture linguistic and structural properties of prompts, and (ii) a set of *semantic or pragmatic smells*, which relate to contextual adequacy, style expectations, and bias. This distinction emerges clearly from both the taxonomy and the quantitative analyses, and has practical consequences for how prompt quality should be assessed in LLM-assisted software engineering. In this section, we discuss such results and provide actionable implications for software engineers, technical writers, and practitioners who routinely craft prompts for LLM-based tools.

A. Syntactic vs. Semantic Prompt Smells

The syntactic smells—grammar, formatting, and readability—represent low-level linguistic issues that can be assessed reliably through established NLP tools. These dimensions show strong mutual correlations and are almost perfectly predictive of the composite Prompt Quality Score (*PQS*). This suggests that they capture a shared underlying

construct related to “surface clarity”: prompts that are grammatically correct, readable, and well structured tend to receive high overall quality scores, and deficiencies in any of these aspects tend to co-occur.

✦ Surface-level fixes improve clarity but not meaning

Improving grammar, structure, and readability does meaningfully increase the overall clarity of a prompt, as reflected in the high correlation among syntactic scores. Tools that automatically refine text (e.g., grammar checkers or readability enhancers) are therefore useful for achieving a clean baseline. However, such improvements do *not* address issues related to missing context, excessive verbosity, formality mismatches, or bias. Practitioners should not mistake syntactic refinement for comprehensive prompt improvement.

In contrast, semantic smells such as contextual adequacy (*RCS*), complexity–length (*CLS*), formality mismatch (*FMS*), and bias (*BDS*) behave very differently. These dimensions exhibit weak or negligible correlations with syntactic properties and show limited explained variance in the OLS models. This indicates that semantic smells capture qualitatively distinct issues: prompts can be syntactically clean yet still context-poor, overly verbose, stylistically misaligned, or implicitly biased. These findings reinforce that high surface quality is *not* a guarantee of high semantic quality, and that the two classes of smells must be evaluated separately.

✦ Semantic smells require human judgment or LLM-based analysis

Because semantic smells show weak relationships with syntactic ones, they cannot be reliably inferred from surface properties. Detecting missing context or ensuring stylistic alignment requires explicit attention to the task, domain, and intended outcome. Practitioners should therefore incorporate semantic review steps—for instance, verifying whether the prompt contains necessary domain information or articulates constraints unambiguously.

B. On the Relation between Prompt Smells

The relationship analysis reveals two important insights. First, syntactic metrics (*G*, *F*, *C*) collectively explain nearly all variance in *PQS*, meaning that *PQS* is essentially a compact representation of these three base properties. Thus, for tasks where surface-level hygiene is the main concern, practitioners may rely on *PQS* as a sufficiently informative indicator.

✦ Complexity–length is partly predictable but still task-specific

Our results indicate that verbosity and cognitive complexity correlate with readability measures, but not fully. Practitioners should aim for concise, focused prompts, but also ensure that essential context is not lost to oversimplification. Length should therefore be managed strategically, not minimized blindly.

Second, the semantic metrics are only weakly explained by the syntactic ones. For example, RCS has an R^2 of approximately 0.15, and FMS and BDS have even lower values. This suggests that semantic smells are largely independent of the textual features captured by existing readability or grammar-oriented tools. Notably, CLS shows moderate predictability ($R^2 \approx 0.48$), driven partly by its structural overlap with readability and length features, but still retains substantial unique variance.

Taken together, these findings indicate that:

- Syntactic metrics are *necessary* but not *sufficient* to characterize prompt quality.
- Semantic smells introduce orthogonal dimensions that require dedicated assessment methods, likely involving LLM-based or semantic-aware analysis.
- Improving surface quality alone does not mitigate deeper issues such as missing context, bias, or formality mismatches.

🔗 Toward automated prompt quality tooling

Our taxonomy and metrics provide the foundation for integrated prompt quality checkers that combine syntactic analysis with semantic evaluation. For practitioners, such tools could serve as automated “linting” mechanisms for prompts, guiding real-time improvements in LLM interactions within software development workflows.

The key insight is that **prompt quality is multi-dimensional**. Syntactic correctness is easy to measure and helpful to improve, but deeper semantic issues require explicit attention and cannot be captured by surface-level metrics alone. Effective prompt engineering, therefore, demands both linguistic hygiene and semantic awareness.

VI. THREATS TO VALIDITY

Construct validity. This study relies on operationalization of inherently qualitative concepts such as “prompt smells” and “prompt quality”. The proposed taxonomy is derived from a rapid review and from the inspection of a specific corpus of prompts, rather than from a fully systematic mapping of the literature and practices. As a consequence, some relevant smell types may be missing or only partially captured. Moreover, the metrics we use to approximate each dimension (in particular RCS, FMS, and BDS, which depend on LLM-based judgments) may introduce measurement noise or bias. The results should therefore be interpreted as evidence about one

possible operationalization of prompt smells, not as a definitive measurement standard.

Internal validity. Our analyses are observational and based on correlations and linear regression models. While these techniques allow us to quantify associations between metrics and smell-related dimensions, they do not support causal claims about how specific properties of prompts influence model behavior. Multicollinearity among metrics, especially within the linguistic quality indicators, may affect the stability and interpretability of individual regression coefficients. In addition, any errors or inconsistencies in the underlying automatic metrics can propagate into the statistical models.

External validity. The dataset used in this study consists of prompts written by developers and shared on public platforms. This population may not be representative of prompts authored in industrial settings, by non-technical users, or in other application domains beyond software development. Furthermore, the analysis focuses on English-language prompts and on a specific family of LLM-based tools and detectors. As a result, the taxonomy and the empirical findings may not generalize to other languages, organizational contexts, or model families without additional validation.

Conclusion validity. The strength of our conclusions is constrained by the size and composition of the dataset, as well as by the choice of statistical models. Linear models with a limited number of observations can be sensitive to outliers and model specification decisions, and R^2 values alone do not fully capture predictive reliability. Therefore, the reported effect sizes and goodness-of-fit measures should be viewed as preliminary indications of which smell dimensions are more amenable to automatic approximation, rather than as definitive evidence of model performance or practical thresholds for tool adoption.

VII. CONCLUSION

This paper presented a preliminary exploration of prompt quality through the identification and measurement of recurring *prompt smells*. By reviewing existing literature and analyzing real-world prompts, we derived an initial taxonomy that distinguishes between syntactic issues—such as grammar, formatting, and readability—and semantic issues related to context, complexity, style, and bias. Our empirical results show that syntactic smells form a coherent cluster, strongly correlated with one another and largely captured by the composite Prompt Quality Score, whereas semantic smells operate independently and cannot be inferred from surface-level text properties. This highlights that prompts may appear well written while still lacking contextual or ethical qualities.

This study represents an early step toward a more principled understanding of prompt quality, and several directions emerge for future work. A natural continuation involves refining and expanding the taxonomy by incorporating additional evidence from industrial guidelines, task-specific prompting practices, and broader literature on human–AI interaction. Equally important is the development of richer semantic metrics capable of capturing intent, task grounding, and pragmatic coherence,

possibly through LLM-based evaluators or embedding-based semantic similarity measures. Another promising avenue is the design of integrated prompt quality assessment tools that combine syntactic and semantic analyses to provide real-time feedback within development environments. Finally, future research should examine how specific prompt smells influence model behavior such as accuracy, hallucination, safety, or adherence to instructions—to understand which issues matter most in practice and how smell mitigation can lead to more reliable LLM-assisted workflows.

Overall, our findings highlight the multidimensional nature of prompt quality and underscore the need for more holistic approaches to prompt evaluation. By laying an initial conceptual and empirical foundation, this work aims to support the development of more robust, transparent, and effective prompting practices in software engineering and beyond.

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